

GARVIT PUJARI

RAG KNOWLEDGE BASE — FINAL

Single-source personal and technical knowledge corpus for the local RAG chatbot.

Prepared from user-provided profile information and the supplied Zenthos and DRDO internship documents.

DOCUMENT PURPOSE

This document is a structured personal knowledge base for a Retrieval-Augmented Generation (RAG) chatbot about Garvit Pujari. It is designed to be chunked, embedded, retrieved, and queried locally. Facts are organized into small, explicit sections so the chatbot can answer questions about Garvit's background, education, skills, internships, projects, and technical work without relying on conversational memory.

PROFILE SUMMARY

Garvit Pujari is a B.Tech Computer Science and Engineering student at Maharaja Agrasen Institute of Technology (MAIT), Delhi, in the 2024–2028 batch. His technical focus includes machine learning, deep learning, computer vision, generative AI, agentic AI, RAG systems, LLM applications, and full-stack AI application development. He has internship/project experience involving video understanding, drone intelligence, computer vision, trajectory prediction, AI interview systems, and environmental monitoring.

PERSONAL INFORMATION

Full name: Garvit Pujari.
Date of birth: 20 May 2006.
Residence: Vaishali, Ghaziabad, Uttar Pradesh.
Email: garvitpujari98@gmail.com.
Contact: 9310723510.
LinkedIn: <https://www.linkedin.com/in/garvit-pujari-919774328/>
LeetCode: <https://leetcode.com/u/garvitpujari/>

EDUCATION

School: Modern School, Vaishali.
Class 10: 96.6%, CBSE.
Class 12: 91.8%, CBSE, Science.
Undergraduate degree: B.Tech in Computer Science and Engineering.
Institute: Maharaja Agrasen Institute of Technology (MAIT), Delhi.
Batch: 2024–2028.
Current CGPA: 9.03.

TECHNICAL SKILLS

Programming and development: Python, C, C++, JavaScript, HTML, CSS, React, FastAPI, Flask, Node.js.
Machine learning and deep learning: supervised learning, computer vision, CNNs, LSTM/GRU, object detection, object tracking, trajectory prediction, PyTorch.
Generative AI and LLMs: RAG, embeddings, vector databases, LangChain, LangGraph, LLM APIs, conversational agents, agentic AI, voice agents.
AI/ML platforms and deployment: Hugging Face, Streamlit, Docker, Kaggle, cloud/GPU-based experimentation.
Computer vision: YOLO-family detectors, ByteTrack, OpenCV, video processing, multi-object tracking.
Data and retrieval: FAISS, sentence-transformers/Hugging Face embeddings, document loaders, text splitting and retrieval pipelines.

INTERNSHIP — ZENTHOS

Organization: Zenthos.
Role: AI/ML Intern.
Type: Paid internship.
Stipend: ₹10,000 per month.
Period: November 2025 – January 2026.
Project: Jewellery Theft Detection and Video Intelligence.

The work focused on AI-based theft detection from video. The project explored spatio-temporal video understanding rather than treating individual frames independently. Technologies and research directions included Python, PyTorch, video modelling, Mamba-family architectures, CNN-based visual processing, and VideoMamba/VideoMambaPro experimentation.

The available progress report records work on MambaVLT, a Mamba-family model with a more LLM-like and complex architecture, while VideoMambaPro + CNN was being prepared for dataset processing and model training. Training of a VideoMambaPro prototype was completed on a Kaggle T4 GPU, producing a .pth model file. A subsequent issue involved the structure of annotations.json not matching the requirements of the CNN component.

Source note: the supplied Zenthos progress report is dated 29 October 2025 and documents the VideoMamba/MambaVLT research and training progress.

INTERNSHIP — DRDO / DRISHTI AI

Organization: Defence Research and Development Organisation (DRDO), Institute for Systems Studies & Analyses (ISSA), Metcalfe House, Delhi.
Project: DRISHTI AI — Drone Recognition, Intelligence, Surveillance, Trajectory & Hazard Identification.
Mentor listed in the report: Ms. Shreeparna, Scientist 'B'.
Report-stated internship period: 10 June 2026 to 1 August 2026. The user has also described this internship as June–July 2026.

DRISHTI is an AI-based drone surveillance and intelligence pipeline designed to move beyond frame-level drone detection toward predictive motion analysis and threat-oriented assessment. The pipeline combines YOLOv11 for drone detection, ByteTrack for persistent multi-object tracking, LSTM-based sequence modelling for future trajectory prediction, and a threat-assessment layer.

The report describes the problem as one where frame-wise detection alone cannot provide movement, trajectory, restricted-zone entry, or suspicious-behaviour intelligence. DRISHTI therefore models the movement history of drones over time.

DRDO / DRISHTI — DATA AND DATASET ENGINEERING

The DRISHTI work involved substantial dataset selection and engineering. DUT Anti-UAV Tracking was selected for detection training because it provided 20 annotated drone video sequences, frame-wise bounding-box labels, and continuous video data suitable for tracking. Additional non-drone COCO samples were added to reduce false positives.

The report states that Anti-UAV was explored but rejected for the final purpose because the drone served as the camera platform rather than the visible target. AU-AIR was also explored because of its rich telemetry, but its visual perspective did not match the requirement: the drone was not the visible object of interest.

The final tracking dataset contained 20 videos, 123 unique track IDs, and 20,799 tracked records. Tracking records included video_id, frame_id, track_id, confidence, bounding-box coordinates, center coordinates, width, and height.

DRDO / DRISHTI — TRAJECTORY FEATURES

Tracked trajectories were converted into a temporal learning dataset. Spatial features included cx, cy, width, and height. Motion features included vx, vy, speed, heading, and acceleration.

The data was grouped by video_id and track_id and sorted by frame_id so that each drone's motion remained temporally continuous. Sliding windows were then constructed for sequence learning.

The main reported multi-step setup used 20 past frames and 9 features per frame, with the next 5 future positions as prediction targets. The report gives a final multi-step dataset of 18,689 sequences with input shape (20, 9) and target shape (10, corresponding to five future (cx, cy) positions).

DRDO / DRISHTI — LSTM MODEL

The trajectory prediction module used a stacked LSTM.

Input size: 9.

Hidden size: 128.

Number of LSTM layers: 2.

Dropout: 0.3.

Output: future trajectory coordinates.

Loss: Mean Squared Error (MSE).

Optimizer: AdamW.

Scheduler: ReduceLROnPlateau.

Batch size: 64.

Maximum epochs: 50.

Early-stopping patience: 8.

Mixed-precision training and gradient clipping were used.

The report states that the retrained model achieved a best validation loss of approximately 0.0042.

DRDO / DRISHTI — THREAT ASSESSMENT

Threat assessment uses outputs from detection, tracking, and trajectory prediction. The considered signals include drone speed, direction of movement, trajectory trend, proximity to restricted or sensitive zones, and predicted future movement.

The system therefore produces a risk-oriented interpretation rather than only a drone/non-drone label. The report describes the final workflow as detection → tracking → trajectory extraction → motion-feature computation → sequence learning → trajectory prediction → threat assessment.

DRDO / DRISHTI — RESULTS

The report states that YOLOv11 + ByteTrack generated persistent drone identities and stable localization across frames. A Hugging Face prototype demonstrated real-time processing with multiple drones. The report also reports 45+ FPS on an RTX 3060 for the detection and tracking pipeline.

The trajectory dataset was built from 20 videos, 123 unique track IDs, and 20,799 tracked records. The multi-step prediction dataset contained 18,689 sequences. The retrained LSTM reached approximately 0.0042 best validation loss.

The overall system produced an end-to-end workflow from raw video to detection, tracking, trajectory prediction, and threat-aware interpretation.

DRDO / DRISHTI — IMPORTANT RESEARCH LESSON

A major learning from the internship was dataset alignment. The AU-AIR experiment technically trained an LSTM, but the predicted trajectories were not satisfactory because the dataset's visual perspective did not match the problem. The drone in AU-AIR acted as the camera platform, while DRISHTI required the drone to be the visible target.

This led to the conclusion that telemetry alone is insufficient for the target problem; the data must represent visible drones with persistent identities and motion histories suitable for future-path prediction.

WAVE CLEANUP — OCEAN WASTE DETECTION AND MAPPING

Wave Cleanup is an environmental computer-vision project focused on detecting ocean waste and using spatial information to support efficient cleanup operations.

The project uses a YOLOv8-based object detector fine-tuned to recognize approximately 40 classes of waste. The system is intended to identify waste hotspots from visual input and map detected waste locations so that cleanup resources can be directed toward areas with higher observed waste concentration.

The project combines object detection with hotspot mapping and live map visualization. The core idea is to move from simple waste detection toward operational environmental intelligence: detect waste, associate detections with locations, visualize the distribution, and support more efficient cleanup planning.

PLACEMENT AI / AI CAREER ASSISTANT

Placement AI is an AI-powered career and interview platform built around multiple specialized modules rather than a single generic chat interface.

The system includes resume analysis, an AI interview agent, coding interviews, technical interviews, and an AI career assistant. The interview agent is designed for live, human-like interaction and can ask questions dynamically, including follow-up questions based on the candidate's previous answers. The system can analyze responses and provide feedback rather than only generating a static list of questions.

The project explores voice-based interaction, interview flow management, real-time conversational behaviour, evaluation/analysis, and career assistance. A key design distinction from a normal ChatGPT conversation is that the system is task-specific and workflow-driven: it controls an interview process, maintains interview context, selects follow-up questions, and evaluates the candidate instead of simply acting as an open-ended general chatbot.

OTHER AI / SOFTWARE PROJECT EXPERIENCE

Garvit has also worked on RAG and LLM applications, including document-based chatbots using document loaders, text splitters, embeddings, vector stores such as FAISS, retrievers, and LLM APIs. He has worked with LangChain and LangGraph for agentic and conversational workflows.

Other project areas include:

Road Intelligence Agent: route-aware AI mobility system combining road-event detection, external road intelligence, conversational follow-up, and voice interaction.

Anti-UAV / drone detection: YOLO-based drone detection using UAV datasets and computer-vision pipelines.

Drone trajectory prediction: LSTM/GRU-based sequence modelling using temporal movement features.

YouTubeMind: a YouTube-video RAG application using a React frontend and FastAPI backend, with transcript extraction, text splitting, Hugging Face embeddings, FAISS retrieval, and an LLM response layer.

Resume Intelligence System: resume processing and matching using TF-IDF, cosine similarity, Flask, and LLM APIs.

AI / COMPUTER VISION WORKFLOW EXPERIENCE

Garvit's computer-vision work commonly follows a pipeline pattern:

1. Acquire or select task-specific visual data.
2. Prepare and validate annotations.
3. Fine-tune or use a pretrained detector.
4. Run inference on images or video.
5. Add tracking when temporal identity is required.
6. Engineer temporal or spatial features.
7. Train a downstream sequence or prediction model when needed.
8. Build an application or visualization layer.
9. Deploy or prototype through platforms such as Hugging Face or Streamlit.

The DRDO DRISHTI project is a representative example: YOLOv11 detection → ByteTrack tracking → trajectory extraction → motion features → LSTM prediction → threat assessment.

GENERATIVE AI / RAG EXPERIENCE

Garvit has practical experience building RAG applications. A typical architecture he has used is:

documents → document loader → text splitter → embeddings → vector database/retriever → prompt construction → LLM → response.

Tools used across these projects include LangChain, LangGraph, Hugging Face embeddings, FAISS, ChatGroq/Groq-based models, FastAPI, Streamlit, and React.

He has also worked on conversational agents where retrieval, tools, memory/context, and structured workflows are used to produce task-specific responses.

DSA / CODING PROFILE

Garvit practices data structures and algorithms using C++ and LeetCode. Topics worked on include arrays, binary search, stacks, queues, trees, recursion, and dynamic programming, with graph practice being comparatively less regular.

LeetCode profile: <https://leetcode.com/u/garvitpujari/>

HACKATHON / EXTRACURRICULAR PROFILE

Garvit has participated in hackathon-oriented AI projects and has worked in team-based project environments. Areas of work include generative AI, agentic AI, computer vision, environmental monitoring, road intelligence, and AI application development.

He has also participated in student/technical activities including the MAIT AWS Students Chapter and has experience organizing/volunteering in events.

CAREER / TECHNICAL INTERESTS

Primary technical interests: machine learning, deep learning, computer vision, generative AI, agentic AI, LLM applications, RAG, multimodal AI, video understanding, and AI-powered real-world systems.

A recurring pattern across Garvit's projects is combining an ML model with an application or decision-support layer. Examples include drone detection plus trajectory prediction and threat assessment, waste detection plus hotspot mapping, and interview intelligence plus a conversational workflow.

SOURCE AND FACT-CONFIDENCE NOTES

Primary uploaded sources used for internship details:

1. DRDO Summer Internship Report — DRISHTI AI. The report explicitly documents the project architecture, datasets, methodology, model configuration, experiments, results, and future scope.
2. Zenthos Progress Report dated 29 October 2025. This source documents MambaVLT/VideoMambaPro research and training progress.

Personal, education, project, and skills information in this knowledge base also incorporates information explicitly supplied by Garvit in the current conversation. Where a project is described from the user's summary rather than a supplied report, the wording is intentionally kept at the level supported by that summary.

Important date note: the DRDO report states 10 June 2026 to 1 August 2026. The user summarized the internship as June–July 2026. The exact report-stated dates are preserved here to avoid silently changing the source.

RAG CHATBOT BEHAVIOR GUIDANCE

When answering questions about Garvit, the chatbot should:

Prefer explicit facts from this document.

Distinguish internship-report facts from user-provided profile facts.

Avoid inventing technologies, metrics, dates, roles, or achievements.

If two sources disagree, state the discrepancy instead of silently choosing one.

For technical questions, explain both the high-level purpose and the concrete implementation when available.

For questions about a project, mention the problem, approach, technologies, data, model, results, and limitations when those facts are available.

If information is not present, say that it is not documented in the current knowledge base.

Treat personal contact information as profile data and avoid unnecessarily exposing it in answers unless the user asks for it.

END OF KNOWLEDGE BASE

This document is intended to be used as the source corpus for Garvit's local RAG chatbot prototype.